Technical Research

CHAPTER: 9

Features	Writing prompt	Purpose	Audience	Sources	Publication format
Academic writing	General topic assigned by the teacher	Communicating what the student knows about the topic, to earn a high grade	The teacher who assigned the project	Secondary sources, for the most part	Academic papers Presentations and posters at academic conferences
Workplace writing	Specific workplace situation, question, or problem raised by the writer or by a supervisor	Providing information needed to answer a question or make a decision	Often several people with differing professional backgrounds	Secondary sources serve as foundation for primary research	Reports Proposals Workplace presentations Presentations and posters at professional conferences

1. Secondary Sources

Information about a topic that has been shared through print, recorded media, or presentations. Secondary sources provide researchers and readers with the background information they need by establishing the professional and intellectual context for an issue or problem.

Secondary Sources

Online Search

- Keyword search
- Author or title search

Library Resources

- Books
- Periodicals
- Research journals
- Newspaper
- Company directories
- Dictionaries, Encyclopedias, and Other General References

2. Conducting **Primary Research**

- Interviews
- Focus groups
- Surveys/Questionnaires
- Laboratory experiments, or field observations
- Original works such as diaries, company reports, and correspondence, as well as documents that are the subject of analysis, such as user's manuals and Web sites.

Quantitative Research

- In technical communication, this step often involves answering questions about how long it takes to perform a task. Technical communicators may also collect and analyze statistics from surveys and interviews. Quantitative research is judged by validity and reliability.
- Research is valid if it measures what it was designed to measure.
- Research is reliable if it can be repeated with the same results.
- Quantitative research collects data that can be represented in numbers.

Qualitative Research

Qualitative research is common in technical communication. Qualitative data cannot be represented in numbers. Instead, qualitative research analyzes words, images, processes, or objects.

Information Gathered Through

- Interviews
- Focus Groups
- Field Observations
- Document Analysis

Using Questionnaire and Interviews

Step 1: Designing a questionnaire:

1. Keeping reader's need while preparing the questionnaire
2. Write a precise purpose statement
3. Limit the number of questions
4. Ask mostly objective questions
5. Include precise and concise instructions at the top of the form
6. Provide clear questions that are easy to answer:

Avoid four common problems:

1. Biased in phrasing
2. Use of undefined terms
3. Use of more than one variable
4. Questions that require too much homework
5. Asking double-barreled questions

Step 1: Preparing for the interview

1. Develop a list of specific objectives for the interview
2. Make clear your main objectives when you contact for the interview
    1. Stress the importance of the person's contribution
    2. Put him or her at ease with your goals and general content of proposed discussion
    3. Set a starting time and approximate length for the interview
3. Prepare and interview outline

It includes:

1. A sequential list of topics you want to cover
2. Specific questions you plan to ask

4. Show that you value your interviewee's time

Showing up early to begin on time/staying on track and ending on time

Step 2: Conducting the Interview

1. Ask open questions
2. Ask closed-ended questions when you need to nail down an answer
3. Use summaries throughout the interview

Results and Analysis

Questionnaire

Themes

Percentages

Statistical Tests (for significance)

Presented through

Tables

Graphs

Interviews

- Presented in the written form (Paragraph/s)